# Car seats (if retained in HMTc scope): HMTc Standard

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CITE AS

Pendergrass, K. (2026). Car seats (if retained in HMTc scope): HMTc Standard. Institute of Contaminant Standards.

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CORRESPONDENCE

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KEYWORDS

heavy metals, food safety, dietary exposure

PUBLICATION HISTORY

This revision 2026-08.

## Abstract

The Heavy Metal Tested & Certified heavy-metal-safety standard for car seats (if retained in HMTc scope) in Infant and Child Durable Goods and Textiles (Ages 0-5), evaluated on total element content of the accessible component material (mg/kg of component; dried non-volatile film for coatings): a content basis, not a migration or extraction basis. 5 of 11 panel analytes publish a numeric limit taken from the strictest applicable government maximum; the remainder are held under mandatory control or a substantiated not-material determination.

## 1 | How every number on this standard is set

HMTc Method v2.0 (government-limit-only): the strictest applicable maximum set by a credible sovereign government for this exact product form in the row's declared native basis; a disclosed read-across from the nearest applicable sovereign cell where the exact cell is unregulated; a substantiated not-material determination where the analyte is a species bounded by a published parent; and a control with a named reference method where neither a sovereign value nor a supportable read-across exists. Never occurrence-derived.

## 2 | Master limit table

All values in ppb (µg/kg), as sold basis. A brand meeting every value qualifies for the mark.

| ANALYTE | TIER | HMTC LIMIT | BASIS |
|---|---|---|---|
| Lead (Pb) | Tier 1 | 90,000 | human-grade read-across (Canada) |
| Cadmium (Cd) | Tier 1 | 75,000 | human-grade read-across () |

| | | | |
|---|---|---|---|
| Total arsenic (tAs) | Tier 2 | 25,000 | human-grade read-across (AU) |
| Inorganic arsenic (iAs) | Tier 1 | ≤ tAs | bounded by tAs (reflex speciation) |
| Mercury, total (tHg) | Tier 1 | 10,000 | human-grade read-across (Canada) |
| Methylmercury (MeHg) | Tier 1 | ≤ tHg | bounded by tHg (reflex speciation) |
| Nickel (Ni) | Tier 2 | 930,000 | human-grade read-across (European Union) |
| Tin (Sn) | Tier 2 | 180,000,000 | human-grade read-across (European Union) |
| Aluminium (Al) | Tier 2 | 28,130,000 | human-grade read-across (European Union) |
| Total chromium (Cr) | Tier 2 | 60,000 | screening ceiling (AU) |
| Hexavalent chromium (Cr-VI) | Tier 2 | 3,000 | human-grade read-across (European Union) |

## 3 | Derivation — where each number comes from

Every cell traces to one named sovereign maximum level (the strictest for the nearest analog), or to an explicit control, not-material, or species-bounded determination. No occurrence statistic selects a certification value.

| ANALYTE | HMTC | SOVEREIGN ANALOG / DETERMINATION |
|---|---|---|
| Lead (Pb) | 90,000 | Canada — human-food analog |
| Cadmium (Cd) | 75,000 | — human-food analog |
| Total arsenic (tAs) | 25,000 | AU — human-food analog |
| Inorganic arsenic (iAs) | ≤ tAs | iAs carries no product-specific sovereign cell; it is a component of and bounded by its parent total tAs (iAs <= tAs), an enforceable ceiling via reflex speciation rather than an empty cell. |
| Mercury, total (tHg) | 10,000 | Canada — human-food analog |

| | | |
|---|---|---|
| Methylmercury (MeHg) | ≤ tHg | MeHg carries no product-specific sovereign cell; it is a component of and bounded by its parent total tHg (MeHg <= tHg), an enforceable ceiling via reflex speciation rather than an empty cell. |
| Nickel (Ni) | 930,000 | European Union — human-food analog |
| Tin (Sn) | 180,000,000 | European Union — human-food analog |
| Aluminium (Al) | 28,130,000 | European Union — human-food analog |
| Total chromium (Cr) | 60,000 | government maximum level |
| Hexavalent chromium (Cr-VI) | 3,000 | European Union — human-food analog |

## 4 | Cite this standard

Pendergrass, K. (2026). *Car seats (if retained in HMTc scope): HMTc standard*. Institute of Contaminant Standards.

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## References

[1] AU — human-food maximum levels
Binds tAs. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[2] Canada — human-food maximum levels
Binds Pb, tHg. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[3] European Union — human-food maximum levels
Binds Ni, Sn, Al, Cr-VI. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

### STATUS AND LICENCE

A normative certification standard, editorially reviewed and version-controlled in-house; every value traces to a named sovereign maximum level, disclosed in the derivation. © 2026 Institute of Contaminant Standards / Heavy Metal Certified. All rights reserved.

### COMPETING INTERESTS

This briefing is published by the Heavy Metal Tested & Certified (HMTc) program, operated by the Institute of Contaminant Standards. Its statements about government regulations, toxicology, and occurrence are sourced to the cited primary and expert-body records and to the independently maintained Heavy Metal Index; the program's own certification thresholds are not used as evidence for any factual claim made here.

### DATA AVAILABILITY

### PUBLICATION HISTORY

The works this reference draws on are listed in full in the references above and on the briefing page at https://heavymetalcertified.com/standards/car-seats. Primary legal instruments and expert-body assessments are named with their public identifiers; the underlying occurrence and toxicology corpus is maintained on the Heavy Metal Index.

This revision 2026-08. The full dated record is maintained at https://heavymetalcertified.com/standards/car-seats.